

Problem 1

Payoffs are listed as (Player 1, Player 2). Nash-equilibrium payoffs are shown in red.

1. Matching pennies (Table 9.22)

		Player 2	
		Heads	Tails
Player 1	Heads	1, -1	-1, 1
	Tails	-1, 1	1, -1

Best responses

- Player 1: BR(Heads) = Heads; BR(Tails) = Tails.
- Player 2: BR(Heads) = Tails; BR(Tails) = Heads.

Results

- Pure-strategy Nash equilibria: none. In every cell, at least one player wants to switch (best responses never line up in the same cell), which is the defining feature of this zero-sum "matching" game.
- Dominant-strategy equilibrium: none — neither player has a strategy that is best regardless of the opponent's choice.
- Pareto-efficient / Pareto-dominant NE: not applicable, since there is no NE. (Every outcome is trivially Pareto efficient here because the game is zero-sum — one player's gain is exactly the other's loss — but that is irrelevant without an equilibrium to rank.)

2. The game of chicken (Table 9.23)

		Player 2	
		S	C
Player 1	S	1, 1	8, 2
	C	2, 8	5, 5

Best responses

- Player 1: BR(S) = C; BR(C) = S.
- Player 2: BR(S) = C; BR(C) = S.

Results

- Pure-strategy Nash equilibria: (S, C) with payoffs (8, 2), and (C, S) with payoffs (2, 8) — one player stays on course while the other swerves.
- Dominant-strategy equilibrium: none. Each player's best strategy depends on what the other does (S is best against C, C is best against S), so neither strategy dominates.
- Pareto-efficient NE: both (8, 2) and (2, 8) are Pareto efficient (no feasible outcome makes both players at least as well off with one strictly better).

- Pareto-dominant NE: none. Player 1 prefers (8, 2) while Player 2 prefers (2, 8) — the two equilibria cannot be ranked, since each favors a different player.

3. The comparative-advantage / invisible-hand game (Table 9.24)

		Player 2	
		P	T
Player 1	P	2, 4	1, 1
	T	5, 5	4, 2

Best responses

- Player 1: $BR(P) = T$; $BR(T) = T \rightarrow T$ strictly dominates P for Player 1 ($5 > 2$ and $4 > 1$).
- Player 2: $BR(P) = P$; $BR(T) = P \rightarrow P$ strictly dominates T for Player 2 ($4 > 1$ and $5 > 2$).

Results

- Pure-strategy Nash equilibrium: (T, P) with payoffs (5, 5) — unique.
- Dominant-strategy equilibrium: (T, P), payoffs (5, 5) — each player's dominant strategy (grow the crop they have a comparative advantage in) yields the equilibrium, illustrating the “invisible hand” result.
- Pareto-efficient NE: (T, P) is Pareto efficient — no outcome gives both players at least 5.
- Pareto-dominant NE: (T, P), trivially, since it is the only equilibrium.

4. The assurance game (Table 9.25)

		Player 2	
		E	L
Player 1	E	4, 4	1, 2
	L	2, 1	3, 3

Best responses

- Player 1: $BR(E) = E$; $BR(L) = L$.
- Player 2: $BR(E) = E$; $BR(L) = L$.

Results

- Pure-strategy Nash equilibria: (E, E) with payoffs (4, 4), and (L, L) with payoffs (3, 3) — the classic “coordinate on early or late planting” structure.
- Dominant-strategy equilibrium: none. Each player wants to match the other's timing, so neither E nor L is unconditionally best.
- Pareto-efficient NE: only (E, E) = (4, 4). The other equilibrium, (L, L) = (3, 3), is Pareto dominated by (E, E), since both players do strictly better there.
- Pareto-dominant NE: (E, E) — it Pareto dominates (L, L), yet (L, L) can still persist as a “poverty trap” equilibrium if farmers cannot coordinate expectations, mirroring the Palampur story in the problem.

5. Public-good game / prisoner's dilemma (Table 9.26)

		Player 2	
		N	P
Player 1	N	2, 2	5, 1
	P	1, 5	3, 3

Best responses

- Player 1: $BR(N) = N$; $BR(P) = N \rightarrow N$ strictly dominates P ($2 > 1$ and $5 > 3$).
- Player 2: $BR(N) = N$; $BR(P) = N \rightarrow N$ strictly dominates P ($2 > 1$ and $5 > 3$).

Results

- Pure-strategy Nash equilibrium: (N, N) with payoffs $(2, 2)$ — unique; both free-ride and no one pays for the streetlight.
- Dominant-strategy equilibrium: (N, N) , payoffs $(2, 2)$ — not paying is each player's dominant strategy regardless of the other's choice.
- Pareto-efficient NE: none. $(N, N) = (2, 2)$ is Pareto dominated by $(P, P) = (3, 3)$, where both would be better off — the hallmark inefficiency of the prisoner's dilemma / public-good under-provision problem.
- Pareto-dominant NE: not applicable — the sole equilibrium is not even Pareto efficient, let alone dominant.

Problem 2

Mona and Felix each choose a concession time ($t_M, t_F \geq 0$) for how long they are willing to wait before giving up on the shared soda bottle (value $v = 10$ to each). Waiting costs 1 unit per minute. Whoever concedes first gets nothing (aside from the waiting cost incurred); the other drinks the bottle. If they concede at the same instant, they split the bottle.

1. Mona's payoff function

Whoever has the shorter concession time gives up first, so the game ends at time $\min(t_M, t_F)$, and that elapsed time is the waiting cost paid by both players.

$u_M(t_M, t_F) =$

- $10 - t_F$ if $t_F < t_M$ (Felix concedes first; Mona wins the bottle and pays only the elapsed waiting cost t_F)
- $5 - t_M$ if $t_F = t_M$ (tie: bottle split equally, $v/2 = 5$, minus the common waiting cost)
- $-t_M$ if $t_F > t_M$ (Mona concedes first; she gets nothing and pays her own waiting cost t_M)

2. Mona's best response, $BR_M(t_F)$

(a) $t_F = 5$

- For $t_M > 5$: $u_M = 10 - 5 = 5$ (constant — the payoff doesn't depend on exactly how long Mona commits, only that it exceeds t_F).
- For $t_M = 5$: $u_M = 5 - 5 = 0$.
- For $t_M < 5$: $u_M = -t_M$, maximized at $t_M = 0$, giving $u_M = 0$.

$BR_M(5) = \text{any } t_M > 5$ (payoff 5). Mona should out-wait Felix; the exact margin doesn't matter.

(b) $t_F = 15$

- For $t_M > 15$: $u_M = 10 - 15 = -5$.

- For $t_M = 15$: $u_M = 5 - 15 = -10$.
- For $t_M < 15$: $u_M = -t_M$, maximized at $t_M = 0$, giving $u_M = 0$.

$BRM(15) = 0$. Outlasting Felix would cost more than the bottle is worth, so Mona's best move is to concede immediately.

(c) $t_F = 10$

- For $t_M > 10$: $u_M = 10 - 10 = 0$.
- For $t_M = 10$: $u_M = 5 - 10 = -5$.
- For $t_M < 10$: $u_M = -t_M$, maximized at $t_M = 0$, giving $u_M = 0$.

$BRM(10) = \{0\} \cup \{t_M : t_M > 10\}$ — immediate concession and any concession time past 10 both yield the same payoff, 0. This is the knife-edge case where winning exactly offsets the waiting cost.

3. Checking Nash equilibrium candidates

By symmetry, Felix's payoff function mirrors Mona's: $u_F = 10 - t_M$ if $t_M < t_F$; $5 - t_F$ if $t_M = t_F$; $-t_F$ if $t_M > t_F$.

(a) $(t_F, t_M) = (5, 6)$

- Mona: $t_M = 6 > t_F = 5$, so $u_M = 10 - 5 = 5$. Since $t_M = 6 > 5$, this is in $BRM(5)$ — Mona is best-responding.
- Felix: $t_F = 5 < t_M = 6$, so $u_F = -t_F = -5$. Felix's best response to $t_M = 6$ would be any $t_F > 6$ (giving $u_F = 4$), not $t_F = 5$. Felix is not best-responding.

Not a Nash equilibrium (*Felix would prefer to wait longer*).

(b) $(t_F, t_M) = (15, 0)$

- Mona: $t_M = 0 < t_F = 15$, so $u_M = -t_M = 0$, matching $BRM(15) = 0$ exactly.
- Felix: $t_F = 15 > t_M = 0$, so $u_F = 10 - t_M = 10$. Given $t_M = 0$, any $t_F > 0$ yields the same payoff of 10, so $t_F = 15$ is a best response.

Nash equilibrium, with payoffs $(u_M, u_F) = (0, 10)$.

(c) $(t_F, t_M) = (10, 100)$

- Mona: $t_M = 100 > t_F = 10$, so $u_M = 10 - 10 = 0$. Since $t_M = 100 > 10$, this is in $BRM(10)$ — Mona is best-responding.
- Felix: $t_F = 10 < t_M = 100$, so $u_F = -t_F = -10$. Felix's best response to $t_M = 100$ would be $t_F = 0$ (giving $u_F = 0$), not $t_F = 10$. Felix is not best-responding.

Not a Nash equilibrium (*Felix is waiting too long given how far ahead Mona has committed — he should concede immediately instead*).